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DEIB

2025

Localozation, Navigatiom and Smart Mobility

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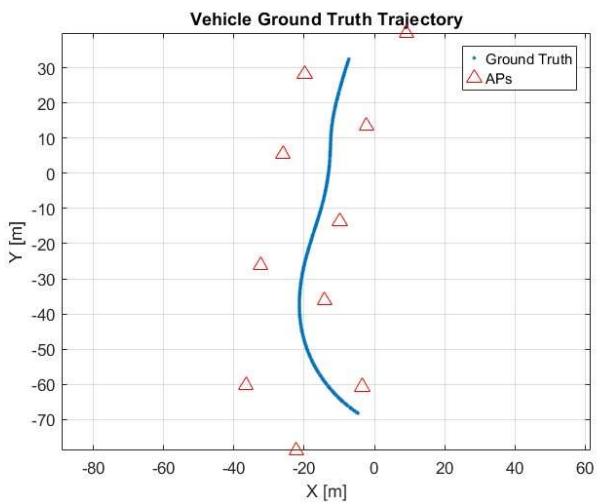
Introduction

- Motivation: Localize a Vehicle in a racetrack.
- Challenge: Measurements are noisy and nonlinear.
- Solution: Use EKF to filter noisy AoA and TDoA measurements.
- Goal: Compare the performance of three methods:
  - AoA-only
  - TDoA-only
  - AoA + TDoA fusion

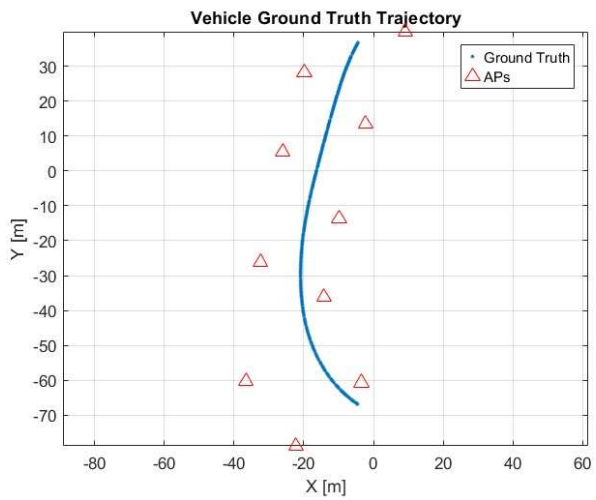
Data Visualization

Firstly, we visualize the ground truth for 3 dataset.

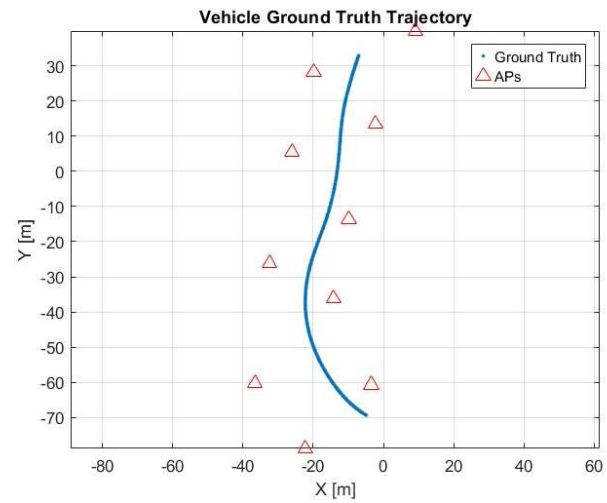
Case 1 : Obstacle



Case 2 : Free-track



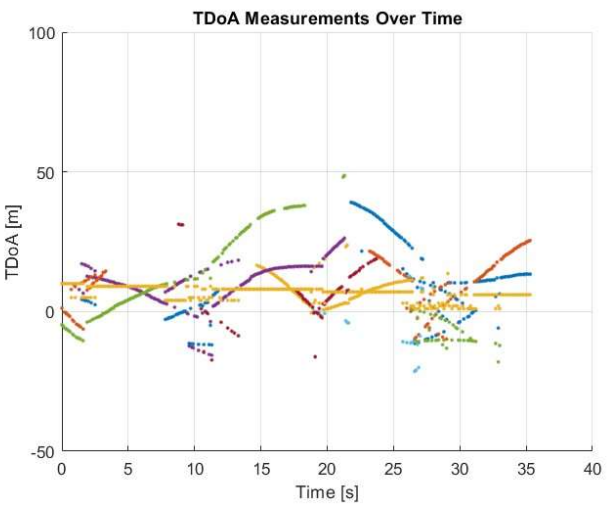
Case 3 : Obstacle



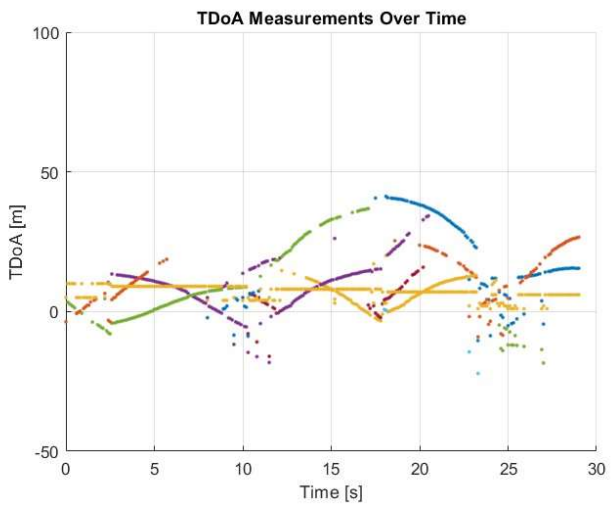
Data Visualization

Secondly, we visualize the TDOA for 3 dataset.

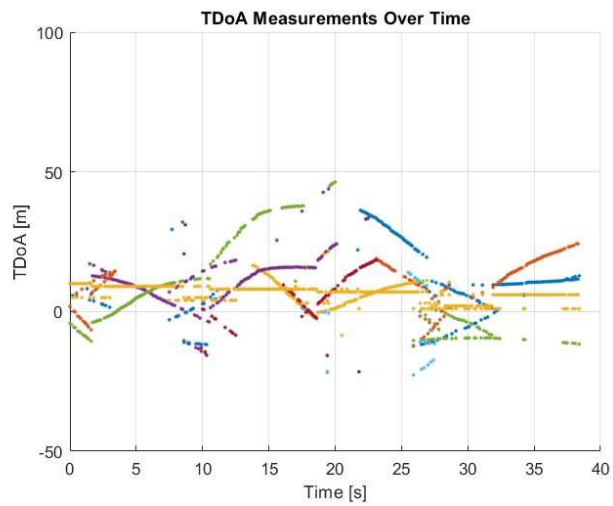
Case 1 : Obstacle



Case 2 : Free-track



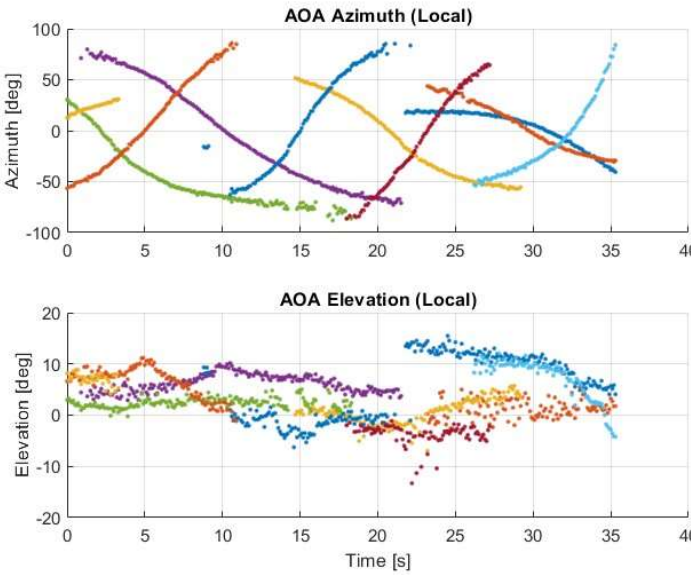
Case 3 : Obstacle



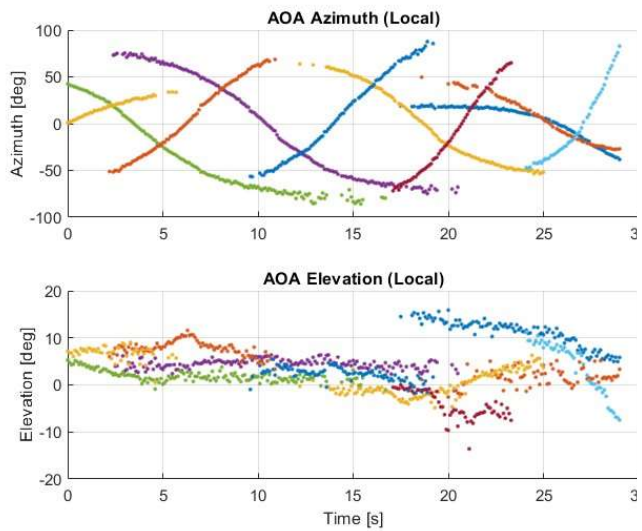
Data Visualization

Thirdly, we visualize the local AOA for 3 dataset.

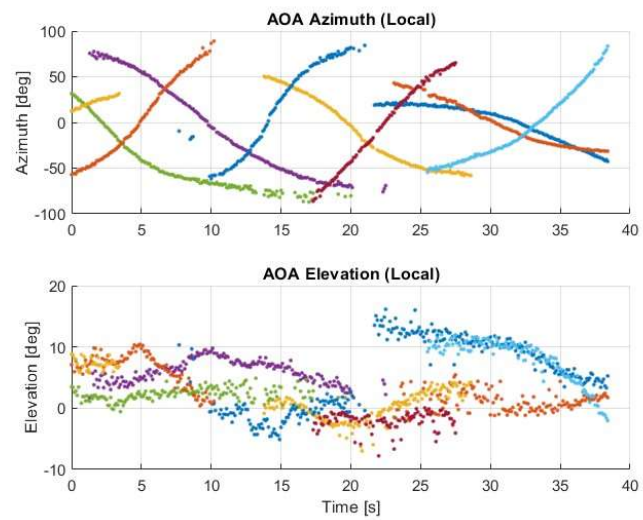
Case 1 : Obstacle



Case 2 : Free-track



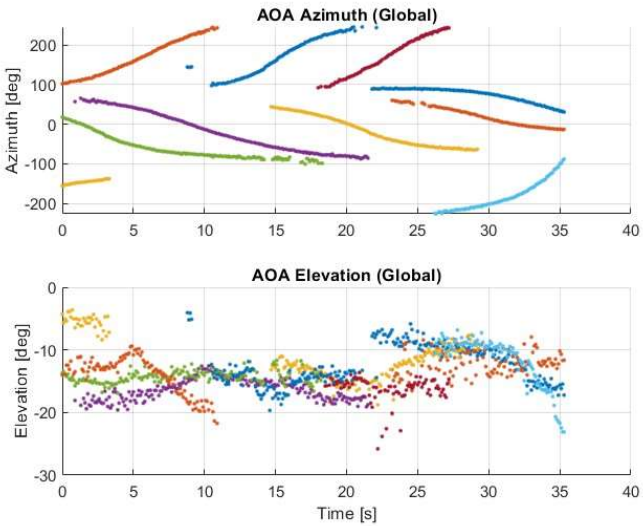
Case 3 : Obstacle



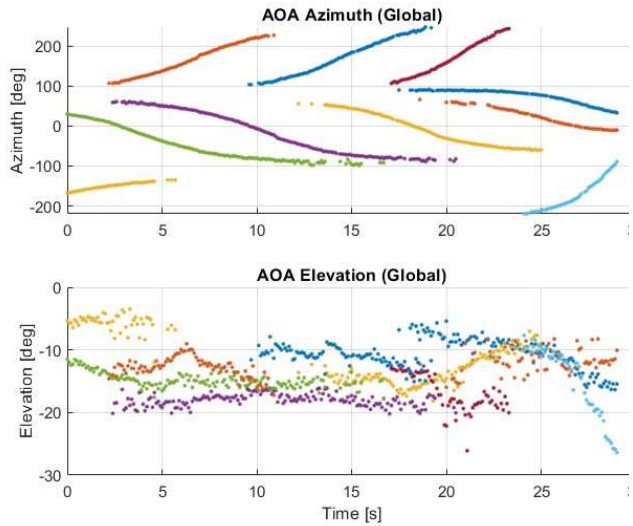
Data Visualization

Finally, we visualize the global AOA for 3 dataset :

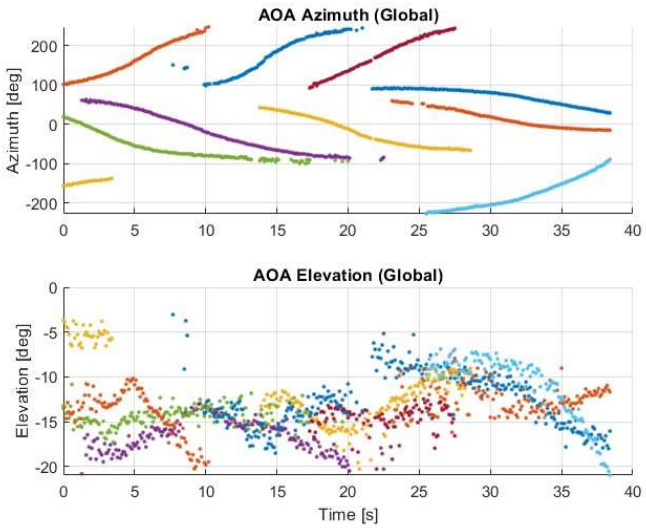
Case 1 : Obstacle



Case 2 : Free-track



Case 3 : Obstacle



EKF setup

Model: Nearly Constant Velocity

Prediction:

$$\hat{\mathbf{x}}_{t|t-1} = \mathbf{F}\hat{\mathbf{x}}_{t-1|t-1}$$
$$\mathbf{C}_{t|t-1} = \mathbf{F}\mathbf{C}_{t-1|t-1}\mathbf{F}^T + \mathbf{Q}$$

Update:

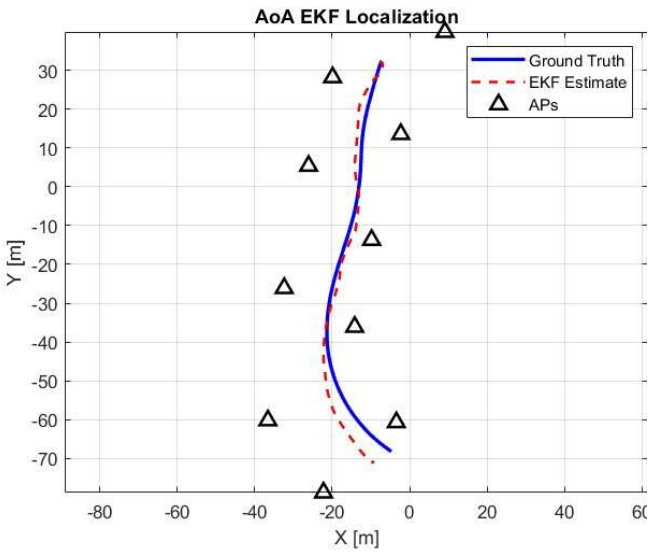
$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \left(\mathbf{C}_{t|t-1}^{-1} + \mathbf{H}_t^T\mathbf{R}_t^{-1}\mathbf{H}_t\right)^{-1}\mathbf{H}_t^T\mathbf{R}_t^{-1}\left(\boldsymbol{\rho}_t - \mathbf{H}_t\hat{\mathbf{x}}_{t|t-1}\right)$$
$$\mathbf{C}_{t|t} = \left(\mathbf{C}_{t|t-1}^{-1} + \mathbf{H}_t^T\mathbf{R}_t^{-1}\mathbf{H}_t\right)^{-1}$$

TDOA	$\Delta d_{i,j} = d_i - d_j = \lvert\mathbf{s}_i - \mathbf{u}\rvert - \lvert\mathbf{s}_j - \mathbf{u}\rvert$	$\left[\frac{s_{jx}-u_x}{d_j} - \frac{s_{ix}-u_x}{d_i}, \frac{s_{jy}-u_y}{d_j} - \frac{s_{iy}-u_y}{d_i}\right]$
AOA	$\alpha_i = \angle(\mathbf{s}_i - \mathbf{u}) = \arctan\frac{s_{iy}-u_y}{s_{ix}-u_x}$	$\left[\frac{s_{iy}-u_y}{d_i^2}, -\frac{s_{ix}-u_x}{d_i^2}\right] = \left[\frac{\sin\alpha_i}{d_i}, -\frac{\cos\alpha_i}{d_i}\right]$

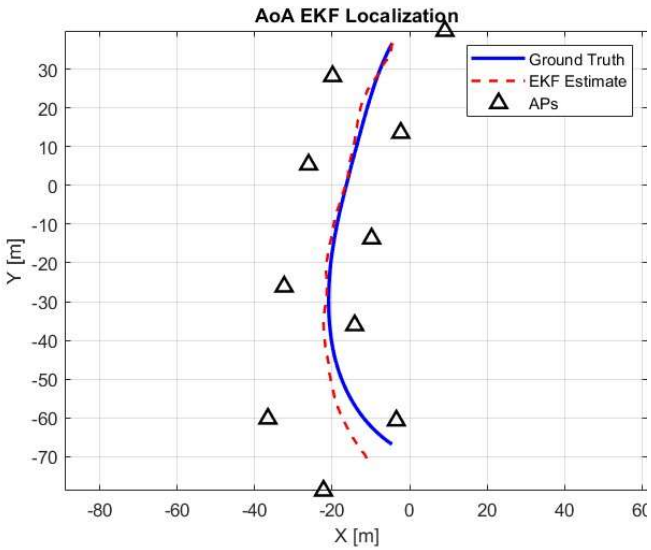
Terms of Jacobian Matrix H

AOA

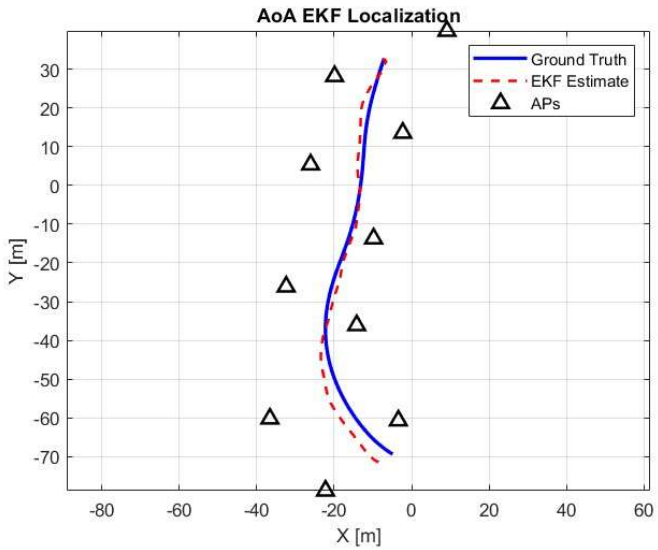
Case 1 : free track



Case 2 : obstacle

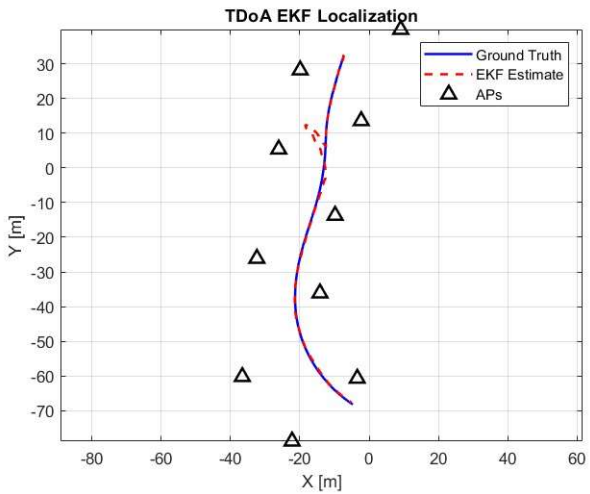


Case 3 : free track

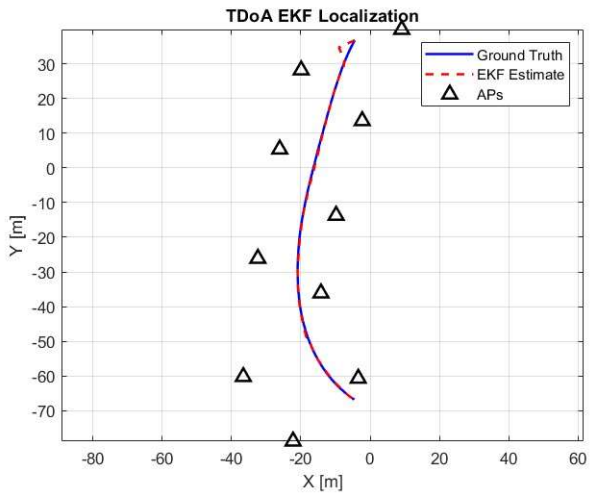


TDOA

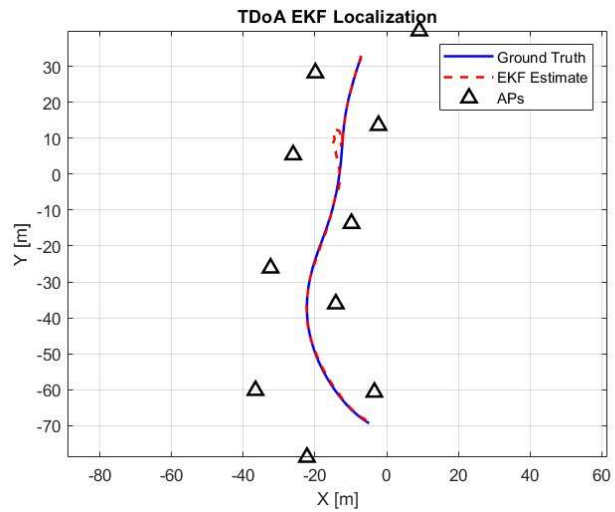
Case 1 : Obstacle



Case 2 : Free-track

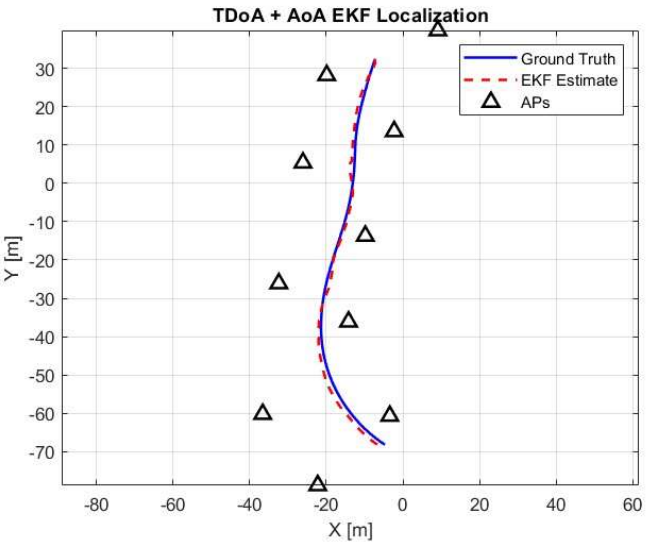


Case 3 : Obstacle

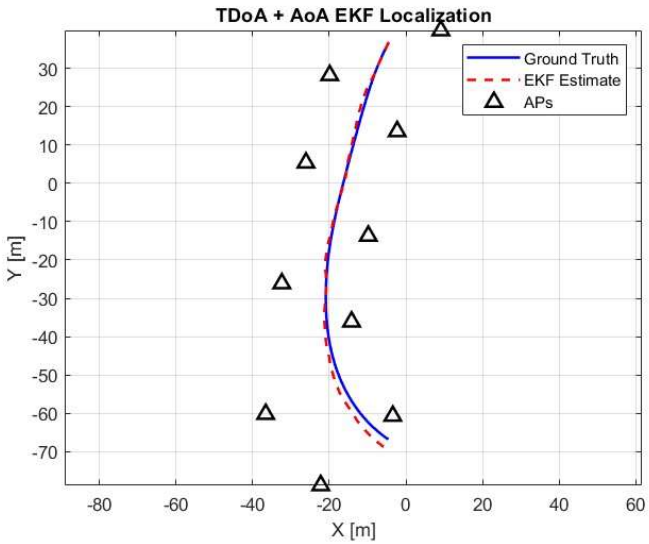


TDOA + AoA

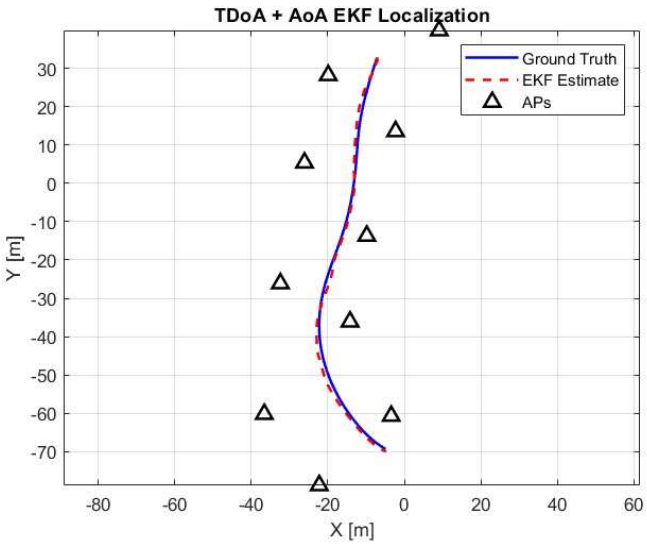
Case 1 : Obstacle



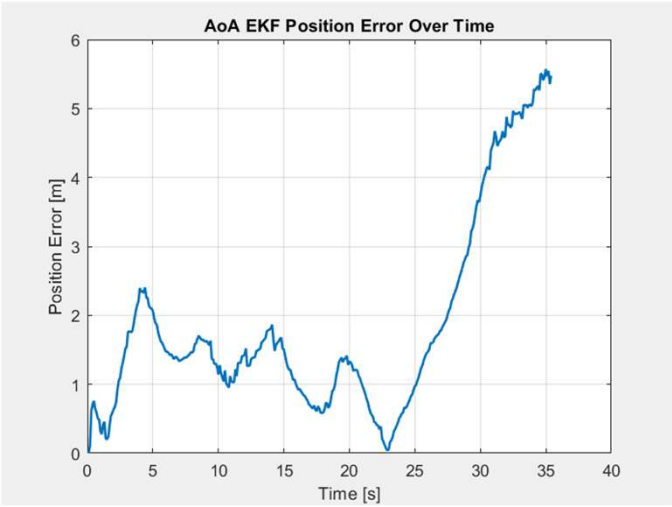
Case 2 : Free-track

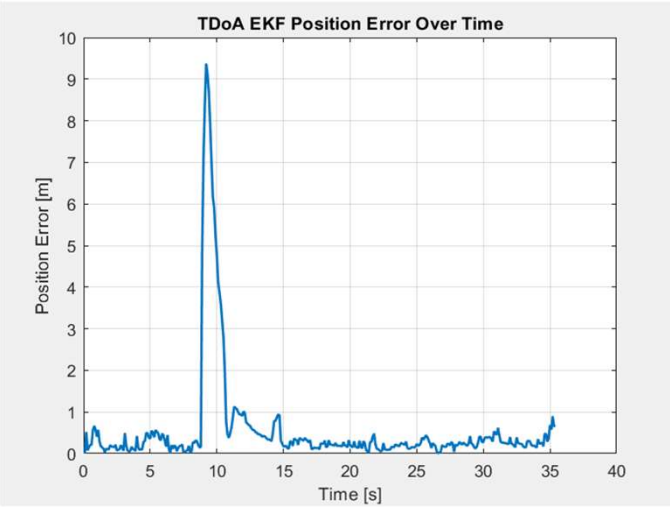


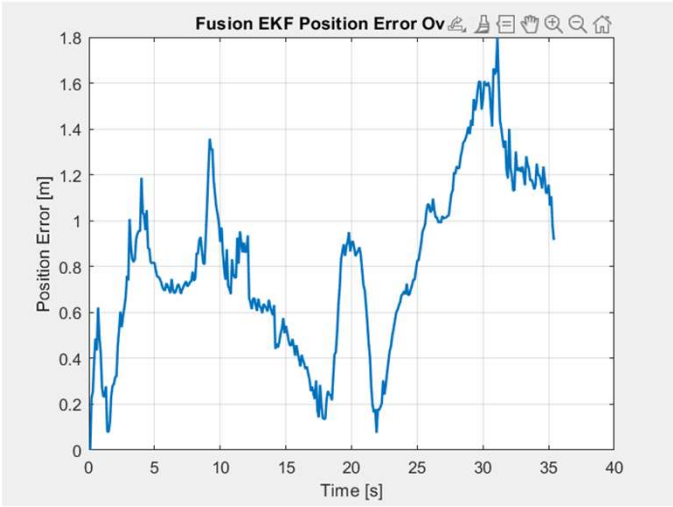
Case 3 : Obstacle



Comparison





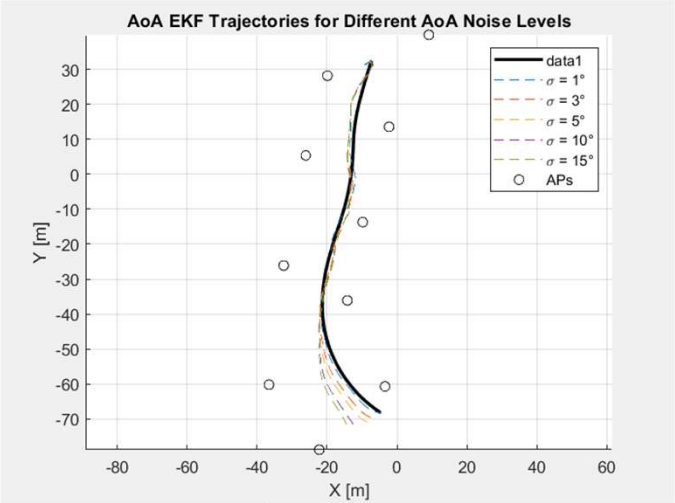


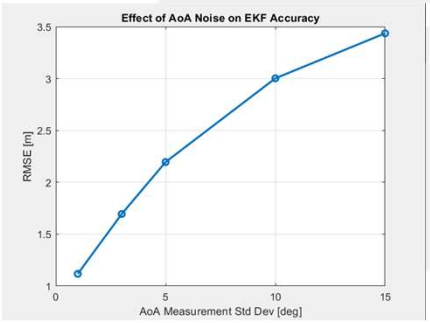

```
AoA EKF RMSE: 2.35 meters
```

```
TDoA EKF RMSE: 1.44 meters
```

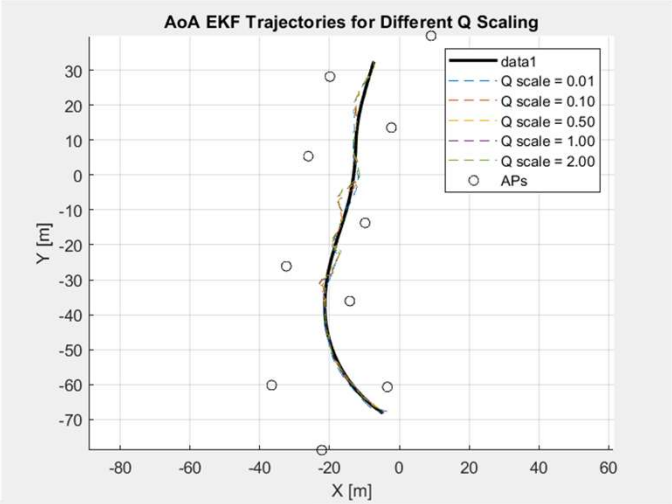
```
Fusion EKF RMSE: 0.89 meters
```

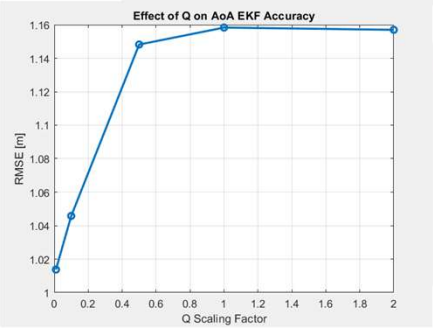
Parameter study: AoA





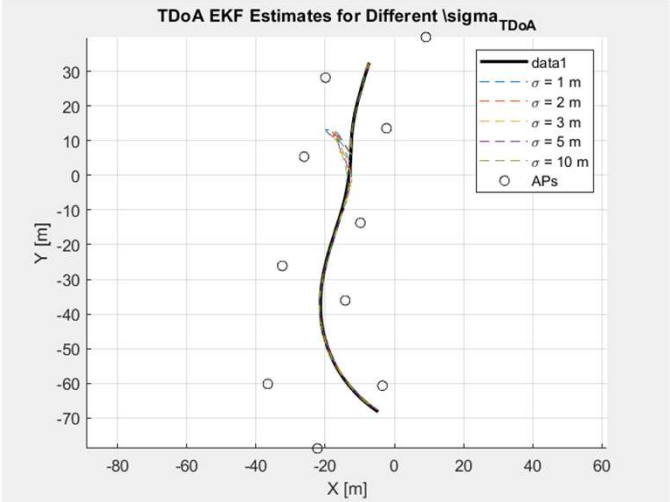

```
--- RMSE vs AoA noise ---
  σ =  1° -> RMSE = 1.12 m
  σ =  3° -> RMSE = 1.69 m
  σ =  5° -> RMSE = 2.19 m
  σ = 10° -> RMSE = 3.00 m
  σ = 15° -> RMSE = 3.44 m
```

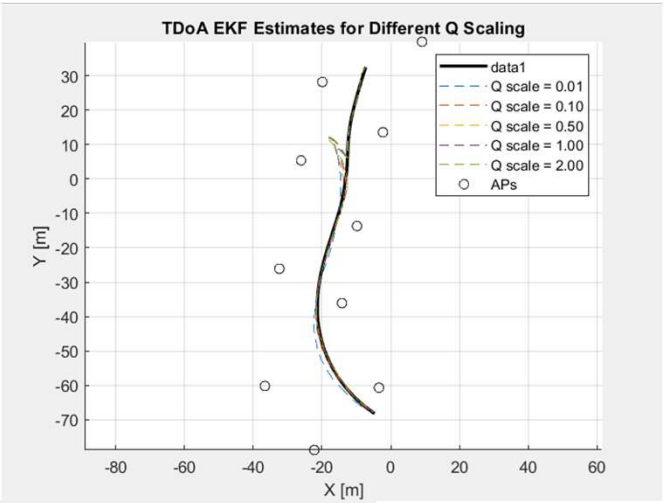


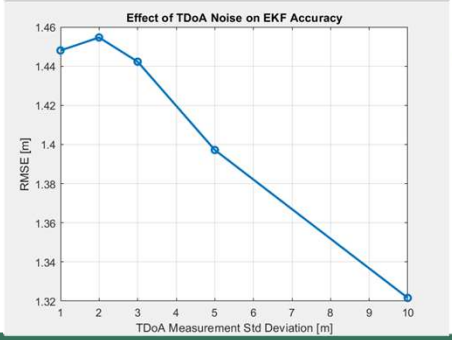



```
--- RMSE vs Q scaling ---
  Q scale = 0.01 -> RMSE = 1.01 m
  Q scale = 0.10 -> RMSE = 1.05 m
  Q scale = 0.50 -> RMSE = 1.15 m
  Q scale = 1.00 -> RMSE = 1.16 m
  Q scale = 2.00 -> RMSE = 1.16 m
```

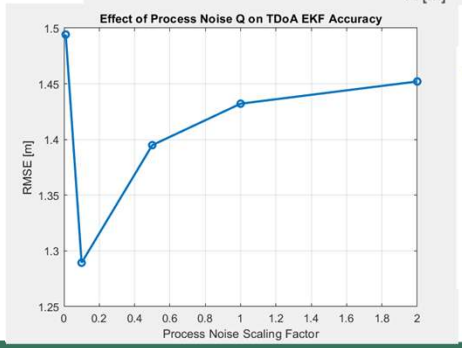
Parameter study: TDoA





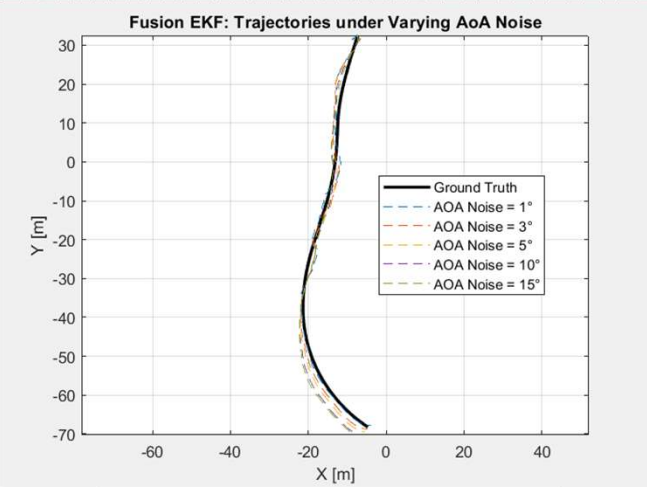


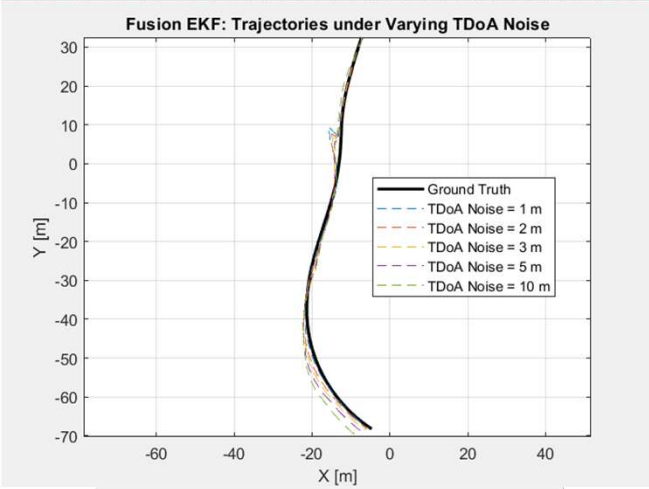

```
--- RMSE vs sigma_tdoa ---
  sigma = 1 m: RMSE = 1.45 m
  sigma = 2 m: RMSE = 1.45 m
  sigma = 3 m: RMSE = 1.44 m
  sigma = 5 m: RMSE = 1.40 m
  sigma = 10 m: RMSE = 1.32 m
```

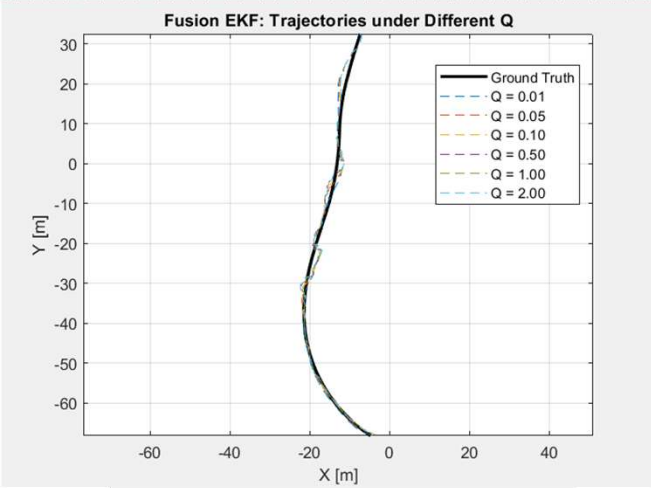


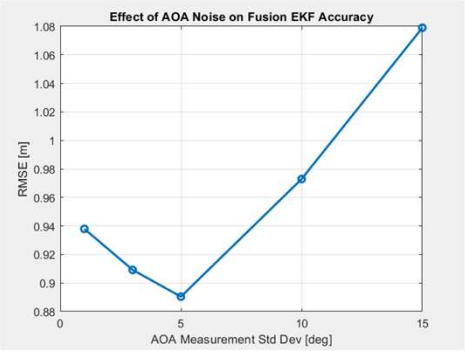

```
--- RMSE vs Q scaling ---
  Q scale = 0.01: RMSE = 1.49 m
  Q scale = 0.10: RMSE = 1.29 m
  Q scale = 0.50: RMSE = 1.39 m
  Q scale = 1.00: RMSE = 1.43 m
  Q scale = 2.00: RMSE = 1.45 m
```

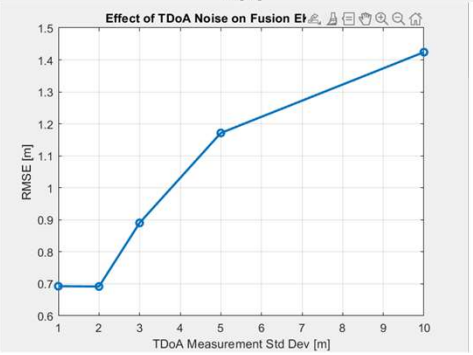
Parameter study: AoA + TDoA

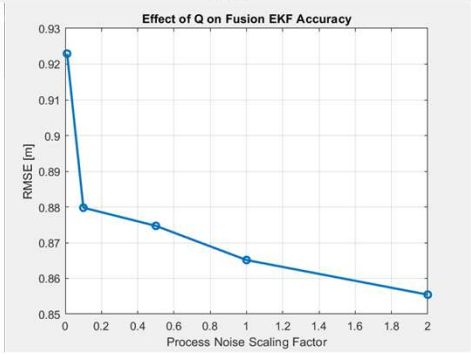












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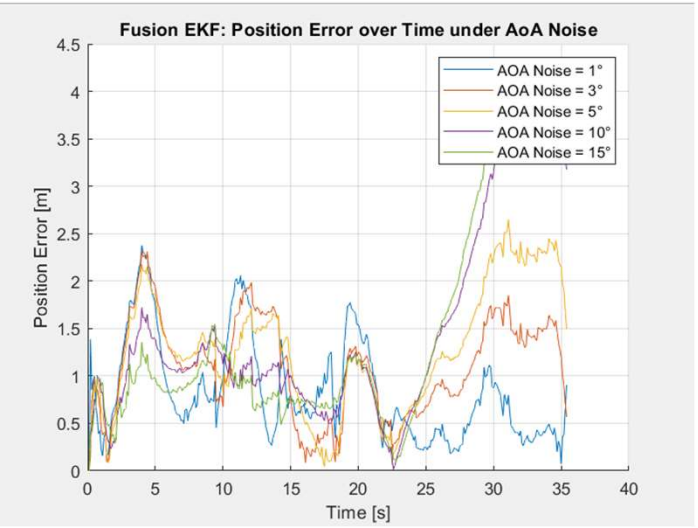
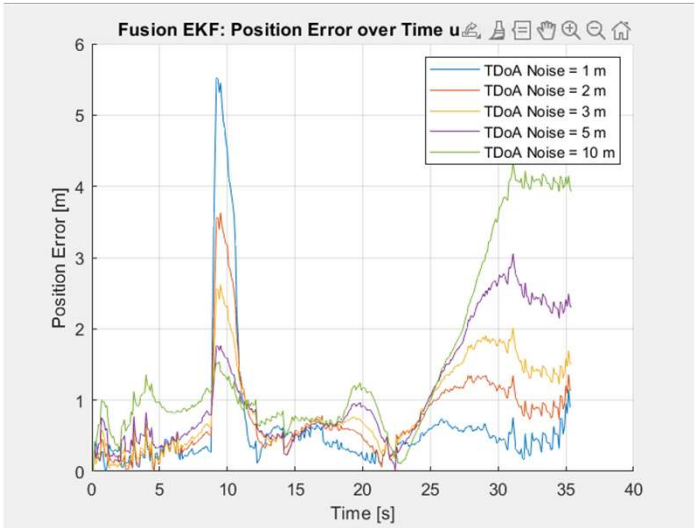
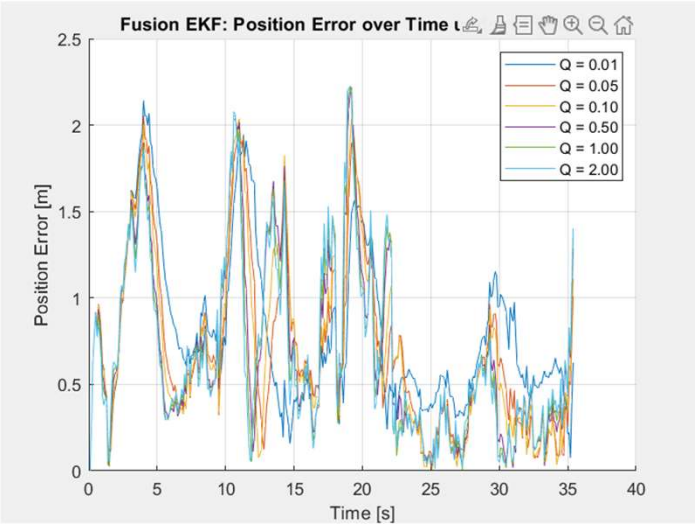
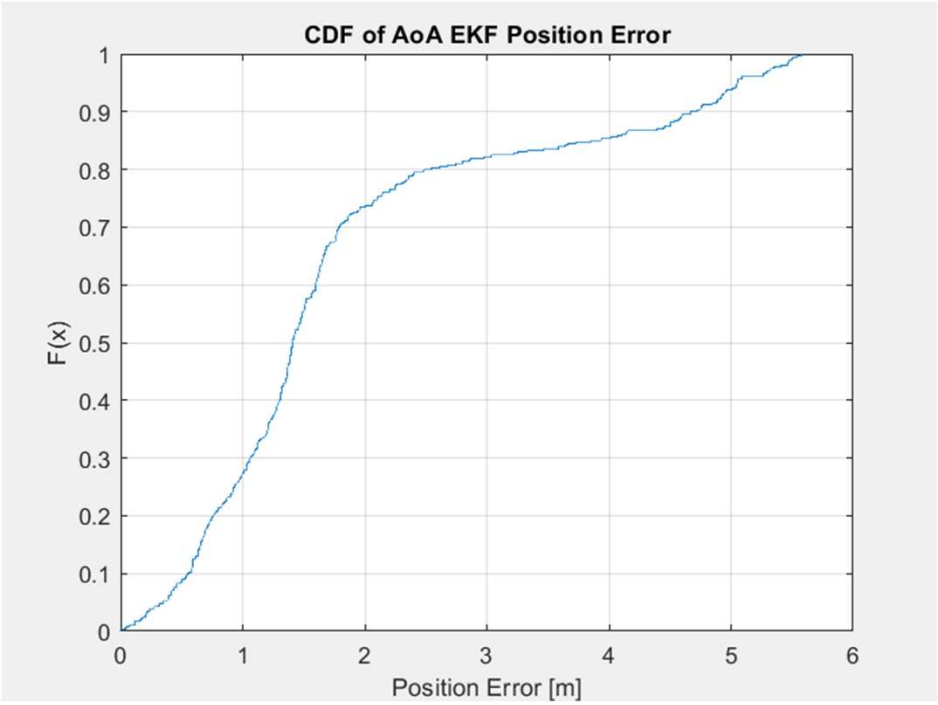
Parameter study: AoA + TDoA

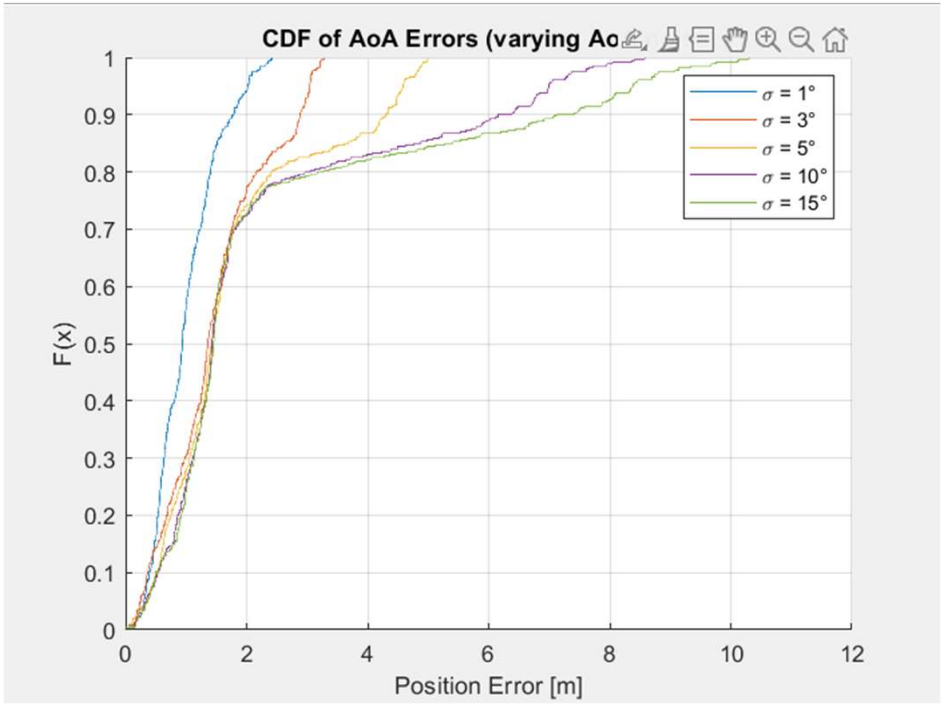




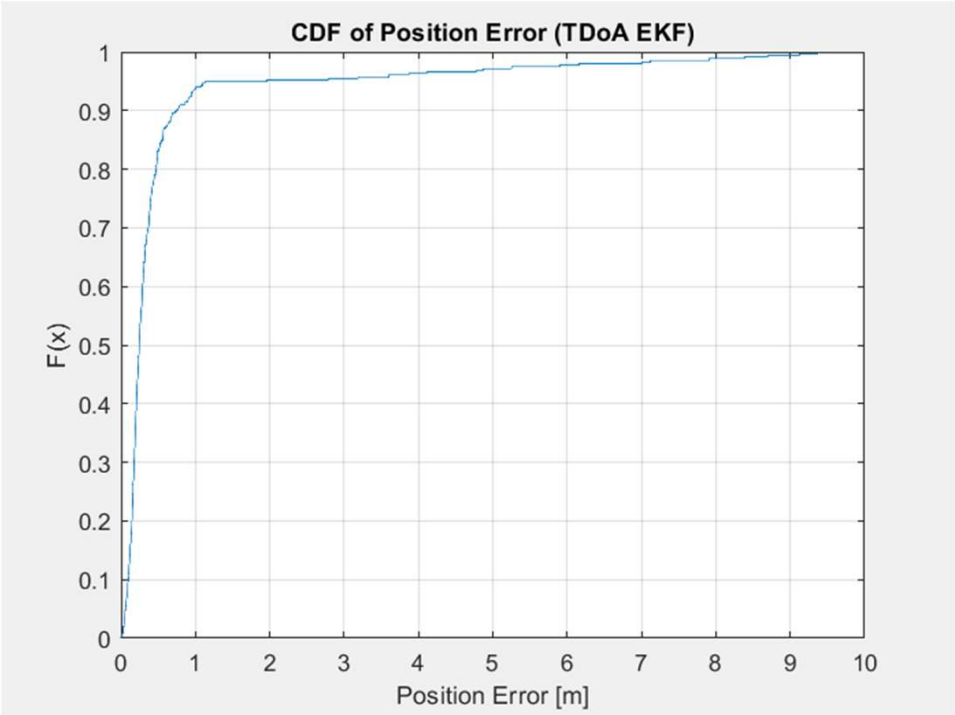


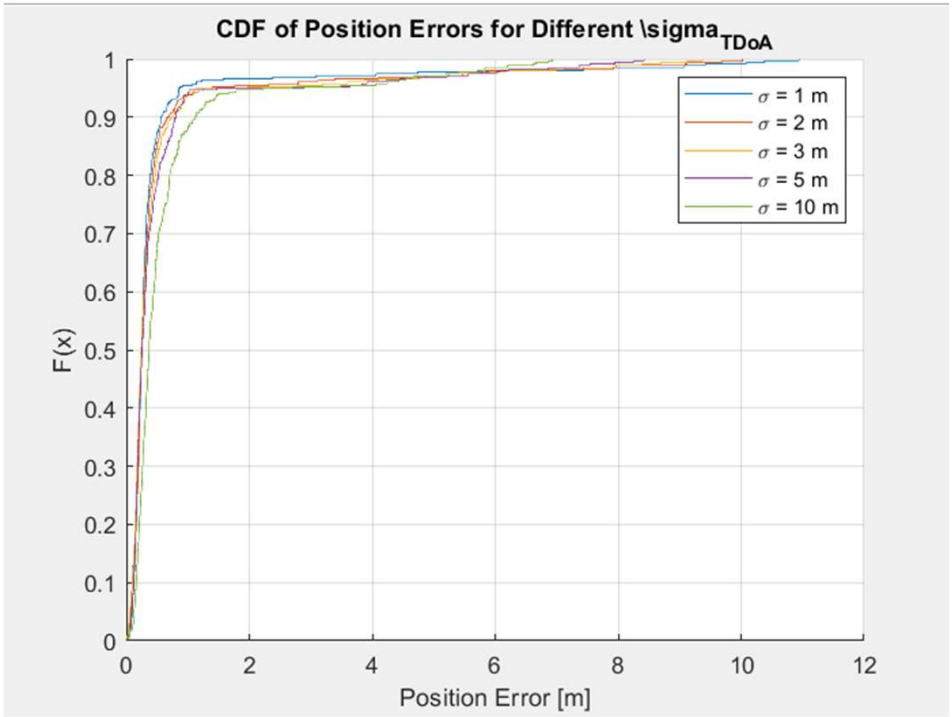
CDF (AoA)


CDF of AoA EKF Position Error
F(x)
Position Error [m]
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
1
2
3
4
5
6

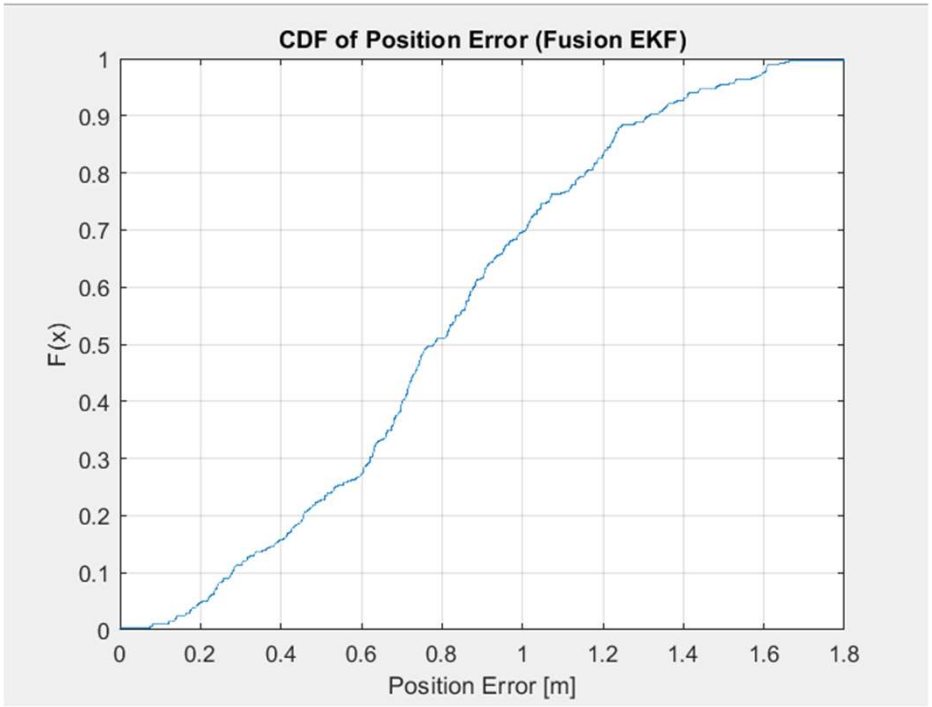

CDF of AoA Errors (varying Ao
F(x)
Position Error [m]
σ = 1°
σ = 3°
σ = 5°
σ = 10°
σ = 15°
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
2
4
6
8
10
12

CDF (TDoA)


CDF of Position Error (TDoA EKF)
F(x)
Position Error [m]
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
1
2
3
4
5
6
7
8
9
10


CDF of Position Errors for Different \sigma_TDoA
F(x)
Position Error [m]
σ = 1 m
σ = 2 m
σ = 3 m
σ = 5 m
σ = 10 m
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
2
4
6
8
10
12

CDF (AoA + TDoA)



Median error: ~0.8 meters.

90% of errors: < **~1.4 meters**.

Maximum error: ~**1.65 meters**.

more **consistent and stable performance**.

Conclusion

- Among the three EKF-based systems, the fusion of AoA and TDoA yielded the lowest RMSE
- The performance of individual systems deteriorates significantly under high noise, while fusion remains relatively robust.
- This project shows that **careful EKF design, tuning, and parameter analysis** can significantly improve localization accuracy
- Future work could explore **nonlinear motion, 3D localization, or deep learning-based sensor fusion** for improved robustness.

Thank you for your time!

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